

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

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# Lines, angles, and triangles

01

10 answers · Rationale-rich · Teach from doc

**Q1****B****B.**  $31^\circ$ 

Choice B is correct. In the figure shown, since  $\overleftrightarrow{QS}$  is a line segment, the sum of the measures of  $\angle PRS$  and  $\angle PRQ$  is  $180^\circ$ . It's given that the measure of  $\angle PRS$  is  $163^\circ$ . Thus, the measure of  $\angle PRQ$  is  $(180 - 163)^\circ$ , or  $17^\circ$ . The sum of the measures of the interior angles of a triangle is  $180^\circ$ . It's given that the measure of  $\angle PQR$  is  $132^\circ$ . Therefore, the measure of  $\angle QPR$  is  $(180 - 17 - 132)^\circ$ , or  $31^\circ$ .

Choice A is incorrect. This is the measure of the supplement of  $\angle PQR$ , not the measure of  $\angle QPR$ .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure of  $\angle PRQ$ , not the measure of  $\angle QPR$ .

**Q2****D****D.** 12

Choice D is correct. It's given that triangle  $ABC$  is similar to triangle  $XYZ$ , where  $A$ ,  $B$ , and  $C$  correspond to  $X$ ,  $Y$ , and  $Z$ , respectively. It follows that side  $AB$  corresponds to side  $XY$  and side  $BC$  corresponds to side  $YZ$ . Since the lengths of corresponding sides in similar triangles are proportional, it follows that  $\frac{XY}{AB} = \frac{YZ}{BC}$ . Substituting 170 for  $AB$ , 60 for  $YZ$ , and 850 for  $BC$  in this equation yields  $\frac{XY}{170} = \frac{60}{850}$ . Multiplying each side of this equation by 170 yields  $XY = 12$ . Therefore, the length of  $\overline{XY}$  is 12.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of  $\overline{YZ}$ , not  $\overline{XY}$ .

**Q3****20**

The correct answer is 20. By the equality given, the three pairs of corresponding sides of the two triangles are in the same proportion. By the side-side-side (SSS) similarity theorem, triangle ABC is similar to triangle DEF. In similar triangles, the measures of corresponding angles are congruent. Since angle BAC corresponds to angle EDF, these two angles are congruent and their measures are equal. It's given that the measure of angle BAC is  $20^\circ$ , so the measure of angle EDF is also  $20^\circ$ .

**Q4****D**

D. No additional information is needed.

Choice D is correct. When two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. It's given that in triangle  $DEF$ , the measure of angle  $D$  is  $47^\circ$  and the measure of angle  $E$  is  $97^\circ$ . It's also given that in triangle  $RST$ , the measure of angle  $R$  is  $47^\circ$  and the measure of angle  $S$  is  $97^\circ$ . It follows that angle  $D$  is congruent to angle  $R$  and that angle  $E$  is congruent to angle  $S$ . Therefore, triangle  $DEF$  is similar to triangle  $RST$  and no additional information is needed.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

## Q5

C

C. 41

Choice C is correct. Vertical angles, which are angles that are opposite each other when two lines intersect, are congruent. The figure shows that lines  $t$  and  $m$  intersect. It follows that the angle with measure  $x^\circ$  and the angle with measure  $y^\circ$  are vertical angles, so  $x = y$ . It's given that  $x = 6k + 13$  and  $y = 8k - 29$ . Substituting  $6k + 13$  for  $x$  and  $8k - 29$  for  $y$  in the equation  $x = y$  yields  $6k + 13 = 8k - 29$ . Subtracting  $6k$  from both sides of this equation yields  $13 = 2k - 29$ . Adding 29 to both sides of this equation yields  $42 = 2k$ , or  $2k = 42$ . Dividing both sides of this equation by 2 yields  $k = 21$ . It's given that lines  $m$  and  $n$  are parallel, and the figure shows that lines  $m$  and  $n$  are intersected by a transversal, line  $t$ . If two parallel lines are intersected by a transversal, then the same-side interior angles are supplementary. It follows that the same-side interior angles with measures  $y^\circ$  and  $z^\circ$  are supplementary, so  $y + z = 180$ . Substituting  $8k - 29$  for  $y$  in this equation yields  $8k - 29 + z = 180$ . Substituting 21 for  $k$  in this equation yields  $8(21) - 29 + z = 180$ , or  $139 + z = 180$ . Subtracting 139 from both sides of this equation yields  $z = 41$ . Therefore, the value of  $z$  is 41.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of  $k$ , not  $z$ .

Choice D is incorrect. This is the value of  $x$  or  $y$ , not  $z$ .

**Q6****C****C.**  $57^\circ$ 

Choice C is correct. It's given that triangle  $CAE$  is similar to triangle  $CBD$ . Corresponding angles in similar triangles have equal measure. Angle  $BCD$  and angle  $ACE$  represent the same angle. It follows that angle  $BCD$  and angle  $ACE$  have equal measure and are corresponding angles. It's given in the figure that angle  $BDC$  and angle  $AEC$  are right angles and therefore have equal measure. It follows that angle  $BDC$  and angle  $AEC$  are corresponding angles. Therefore, angle  $CBD$  and angle  $CAE$  are corresponding angles and have equal measure. It's given that the measure of angle  $CBD$  is  $57^\circ$ , so the measure of angle  $CAE$  is  $57^\circ$ .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

**Q7****B****B.**  $53^\circ$ 

Choice B is correct. It's given that triangle  $LMN$  is similar to triangle  $PQR$ . Corresponding angles of similar triangles are congruent. Since angle  $M$  and angle  $Q$  correspond to each other, they must be congruent. Therefore, if the measure of angle  $M$  is  $53^\circ$ , then the measure of angle  $Q$  is also  $53^\circ$ .

Choice A is incorrect and may result from concluding that angle  $M$  and angle  $Q$  are complementary rather than congruent.

Choice C is incorrect and may result from concluding that angle  $M$  and angle  $Q$  are supplementary rather than congruent.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q8****A**

A.  $\frac{350}{15}$

Choice A is correct. The figure shows that angle  $P$  in  $\triangle QPR$  and angle  $T$  in  $\triangle STR$  are right angles. It follows that angle  $P$  is congruent to angle  $T$ . The figure also shows that the measures of angle  $QRP$  and angle  $SRT$  are both  $x^\circ$ . Therefore, angle  $QRP$  is congruent to angle  $SRT$ . It's given that  $\triangle QPR$  is similar to  $\triangle STR$ . Since angle  $P$  is congruent to angle  $T$ , and angle  $QRP$  is congruent to angle  $SRT$ , it follows that  $\bar{QR}$  corresponds to  $\bar{SR}$ , and  $\bar{QP}$  corresponds to  $\bar{ST}$ . Since corresponding sides of similar triangles are proportional, it follows that  $\frac{SR}{QR} = \frac{ST}{QP}$ . It's also given that the lengths of  $\bar{ST}$ ,  $\bar{QP}$ , and  $\bar{QR}$  are 14, 15, and 25, respectively. Substituting 14 for  $ST$ , 15 for  $QP$ , and 25 for  $QR$  in the equation  $\frac{SR}{QR} = \frac{ST}{QP}$  yields  $\frac{SR}{25} = \frac{14}{15}$ . Multiplying each side of this equation by 25 yields  $SR = \frac{14}{15}25$ , or  $SR = \frac{350}{15}$ . Thus, the length of  $\bar{SR}$  is  $\frac{350}{15}$ .

Choice B is incorrect. This is the result of solving the equation  $\frac{SR}{25} = \frac{14}{20}$ , not  $\frac{SR}{25} = \frac{14}{15}$ .

Choice C is incorrect. This is the result of solving the equation  $\frac{SR}{14} = \frac{15}{20}$ , not  $\frac{SR}{25} = \frac{14}{15}$ .

Choice D is incorrect. This is the result of solving the equation  $\frac{SR}{14} = \frac{15}{25}$ , not  $\frac{SR}{25} = \frac{14}{15}$ .

**Q9****63**

The correct answer is 63. It's given that triangles  $ABC$  and  $DEF$  are congruent with angle  $A$  corresponding to angle  $D$ . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle  $A$  is  $27^\circ$ . It follows that the measure of angle  $D$  is also  $27^\circ$ . It's given that angle  $E$  is a right angle. Therefore, the measure of angle  $E$  is  $90^\circ$ . Let  $x$  represent the measure, in degrees, of angle  $F$ . Since the measures of the angles in a triangle sum to  $180^\circ$ , it follows that  $27 + 90 + x = 180$ , or  $117 + x = 180$ . Subtracting 117 from both sides of this equation yields  $x = 63$ . Therefore, the measure, in degrees, of angle  $F$  is 63.

## Q10

57

The correct answer is 57. Based on the figure, the angle with measure  $y^\circ$  and the angle vertical to the angle with measure  $58^\circ$  are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure  $58^\circ$  also has measure  $58^\circ$ . It's given that lines  $q$  and  $r$  are parallel. Therefore, same side interior angles between lines  $q$  and  $r$  are supplementary. It follows that  $y + 58 = 180$ . If  $y = 2x + 8$ , then the value of  $x$  can be found by substituting  $2x + 8$  for  $y$  in the equation  $y + 58 = 180$ , which yields  $2x + 8 + 58 = 180$ , or  $2x + 66 = 180$ . Subtracting 66 from both sides of this equation yields  $2x = 114$ . Dividing both sides of this equation by 2 yields  $x = 57$ . Thus, if  $y = 2x + 8$ , the value of  $x$  is 57.

### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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